

Alterations in the Treatment of Castration Wounds in Piglets

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ABSTRACT

This study aimed to evaluate the efficacy of Norway spruce ointments on wound healing of castration wounds in piglets. This study included 95 male pigs randomly divided into five treatment groups: Norway spruce balm (Vulpuran), Norway spruce resin (Abilar), pork lard (ointment base of Vulpuran), no treatment (negative control) and antibiotic blue spray (Cyclo spray, positive control). Wound healing parameters (such as healing time, wound size, reddening of wound edges and surrounding, swelling, secretion and wound contamination), microbiological status and the haptoglobin level as an inflammation parameter were investigated. In the Norway spruce groups, some positive effects on wound healing parameters were found. In the first 6 days of treatment, Abilar or Vulpuran showed the smallest means of wound areas, and at the end of the study (day 15 + 17), the highest rates of completely closed wounds compared to the other groups. Vulpuran treatment led to significantly lower wound secretion ($p = 0.003$) and wound contamination ($p = 0.015$) than the untreated control did. Furthermore, the microbiological status was determined using MALDI-TOF-MS and partial 16S rRNA gene sequencing at different days of treatment. A comparison of the five treatment groups on day 3 revealed that Norway spruce led to the lowest rate of wounds colonised with fungi, mainly classified into genus *Candida*, (Abilar 77%, Vulpuran 70%) in comparison with blue spray (89%), lard (100%) and untreated control (100%). Fungi could only be detected in one of the 13 samples treated with Vulpuran on day 8, which nearly reached significance ($p = 0.055$).